Name: _____.

Student number: _____.

Instructions

- Turn off all the electronic devices.
- This is a closed book, and no notes exam.
- Don't use the back of the pages. Ask for extra paper.
- The last page of this exam contains a notesheet that you can tear off.
- You can use any fact in the notesheet, but you must reference which fact you are using.
- You can use every theorem proved in class, but you need to state the theorem.
- Unless otherwise stated, you must justify your answers.
- If you have a question, raise your hand and I will come to you.
- You have 1 hour and 50 minutes to complete the exam.
- Good luck!

Question:	1	2	3	4	5	6	Total
Points:	10	10	10	10	10	10	60
Score:							

1. (10 points) Prove or disprove the following statments.

   (a) If A and B are sets, then $A \setminus (A \setminus B) \subseteq B$.

This statement is True

Assume $x \in A \setminus (A \setminus B)$. Then $x \in A$ and $x \notin A \setminus B$.

Now, $x \notin A \setminus B$ means that either $x \notin A$ or $x \in B$.

Since $x \in A$, it must be $x \in B$.

Thus $A \setminus (A \setminus B) \subseteq B$.

   (b) Let $a \in \mathbb{Z}$. Then $3a \equiv 0 \pmod 6$ if and only if a is even.

This statement is True. We prove both implications.

[$\Rightarrow$] Assume $3a \equiv 0 \pmod 6$. Then $6 \mid 3a$, so $3a = 6k$ for some $k \in \mathbb{Z}$. But, since $6 = 2 \cdot 3$, multiplying both sides of $3a = 6k$ by 3^{-1} we get $a = 2k$ with $k \in \mathbb{Z}$. Hence a even.

[$\Leftarrow$] Assume a even. So there is some $j \in \mathbb{Z}$ such that $a = 2j$. Multiplying by 3, we get $3a = 6j$. Hence $6 \mid (3a - 0)$, which means $3a \equiv 0 \pmod 6$.

□

2. (10 points) Let $f: A \to B$ and $g: C \to D$ be functions. Let $h: A \times C \to B \times D$ defined by

$$h(x, y) = (f(x), g(y)) \qquad \text{for all } x \in A \text{ and } y \in C.$$

(a) Prove that if f and g are injective, then h is injective.

Assume that $(x_1, y_1), (x_2, y_2) \in A \times C$ are such that $h(x_1, y_1) = h(x_2, y_2)$. By definition of h, this says that $(f(x_1), g(y_1)) = (f(x_2), g(y_2))$. Hence $f(x_1) = f(x_2)$ and $g(y_1) = g(y_2)$. Since both f and g are injective, it follows that $x_1 = x_2$ and $y_1 = y_2$. Thus $(x_1, y_1) = (x_2, y_2)$. This proves that h is injective $\square$

(b) Prove that if f and g are surjective, then h is surjective.

Take an arbitrary $(b, d) \in B \times D$. So $b \in B$ and $d \in D$.
Since $f: A \to B$ is surjective, there exists some $a \in A$ such that $f(a) = b$.
Since $g: C \to D$ is surjective, there exists some $c \in C$ such that $g(c) = d$.
Then $h(a, c) = (f(a), g(c)) = (b, d)$.
Thus h is surjective.
$\square$

3. (10 points) Consider the universal set U of all function from $\mathbb{N}$ to $\mathbb{N}$. For each $m \in \mathbb{N}$, consider the set $F_m = \{f \in U \mid f(1) = m\}$.

   (a) Find $\bigcup_{m\in\mathbb{N}} F_m$.

We claim that $\bigcup_{m\in\mathbb{N}} F_m = U$.

We prove both inclusions.

The inclusion [$\subseteq$] follows for free because U is the universe.

[$\supseteq$] Let $f: \mathbb{N} \to \mathbb{N}$ an arbitrary element of U.

If we denote $k = f(1) \in \mathbb{N}$, we have $f \in F_k$ by definition of F_k. Since $F_k \subseteq \bigcup_{m\in\mathbb{N}} F_m$, it follows that $f \in \bigcup_{m\in\mathbb{N}} F_m$.

   (b) Find $\bigcap_{m\in\mathbb{N}} F_m$.

We claim that $\bigcap_{m\in\mathbb{N}} F_m = \emptyset$. Assume for a contradiction that there is some $f \in \bigcap_{m\in\mathbb{N}} F_m$. Thus $f \in F_m$ for all $m \in \mathbb{N}$.

In particular (for $m=1$) we have $f \in F_1$, which means $f(1)=1$

But also (for $m=2$) we have $f \in F_2$, which says that $f(1)=2$

Since f is a function, $f(1)$ cannot have two values.

Thus such f cannot exist.

This proves that $\bigcap_{m\in\mathbb{N}} F_m = \emptyset$ □

4. (10 points) Let A and B be nonempty sets and $f: A \to B$ a function.

(a) Let $U \subseteq A$ and $V \subseteq A$. Prove that, if f is injective, then $f(U \setminus V) = f(U) \setminus f(V)$.

Assume f is injective. We prove both inclusions of $f(U \setminus V) = f(U) \setminus f(V)$.

[$\subseteq$] Let $b \in f(U \setminus V)$. Then there exists some $a \in U \setminus V$ such that $b = f(a)$. Since $a \in U \setminus V$, in particular, $a \in U$. Then $b = f(a) \in f(U)$.

We still need to show that $b \notin f(V)$. Assume for a contradiction that $b \in f(V)$. Then $b = f(a')$ for some $a' \in V$. But then $f(a) = b = f(a')$, so $a = a'$ because f is injective.

But recall that $a \in U \setminus V$, so $a \notin V$. But we have $a = a' \in V$

So $a \notin V$ and $a' \in V$, a contradiction. Hence $b \notin f(V)$.

As a consequence, $b \in f(U) \setminus f(V)$, and [$\subseteq$] is proved.

[$\supseteq$] This is Problem (5)(b) in Practice Final A $\square$

(b) Prove that if $f(U \setminus V) = f(U) \setminus f(V)$ for every $U \subseteq A$ and every $V \subseteq A$, then f is injective.

We prove the contrapositive. Assume that f is <u>not</u> injective. We need to find <u>some</u> $U \subseteq A$ and <u>some</u> $V \subseteq A$ such that $f(U \setminus V) \neq f(U) \setminus f(V)$.

Since f is not injective, there exist $a, a' \in A$ with $a \neq a'$ but $f(a) = f(a')$.

Consider the subsets $U = \{a\}$ and $V = \{a'\}$. Denote $b = f(a) = f(a')$.

Since $a \neq a'$, we have $U \setminus V = \{a\} \setminus \{a'\} = \{a\}$. Thus $f(U \setminus V) = f(\{a\}) = \{b\}$

On the other hand, $f(U) = f(\{a\}) = \{b\}$ and also $f(V) = f(\{a'\}) = \{b\}$

Thus $f(U) \setminus f(V) = \{b\} \setminus \{b\} = \emptyset$.

Since $f(U \setminus V) = \{b\}$ and $f(U) \setminus f(V) = \emptyset$, we have

$f(U \setminus V) \neq f(U) \setminus f(V)$, as desired $\square$

5. (10 points) Let F be the set of all functions $f: \mathbb{R} \to \mathbb{R}$. Define a relation $\sim$ on F by $f \sim g$ if and only if there exists some $c \in \mathbb{R}$ with $c \neq 0$ such that $f(x) = cg(x)$ for all $x \in \mathbb{R}$.

   (a) Show that $\sim$ is an equivalence relation.

Reflexive: Let $f \in F$. Then, for the multiplicative identity $1 \neq 0$ in $\mathbb{R}$, we have $f(x) = 1 \cdot f(x)$ $\forall x \in \mathbb{R}$. Hence $f \sim f$.

Symmetric: Assume $f \sim g$. Then there exists $c \neq 0$ in $\mathbb{R}$ such that $f(x) = cg(x)$ $\forall x \in \mathbb{R}$. Since $c \neq 0$, we have $c^{-1} \in \mathbb{R}$ which is also nonzero. (because $c^{-1} = 0 \Rightarrow 1 = cc^{-1} = c \cdot 0 = 0$). Multiplying both sides of $f(x) = cg(x)$ by c^{-1}, we get $g(x) = c^{-1}f(x)$ $\forall x \in \mathbb{R}$. Hence $g \sim f$.

Transitive: Assume $f \sim g$ and $g \sim h$. Let $c, d \in \mathbb{R}$ both nonzero such that $f(x) = cg(x)$ and $g(x) = dh(x)$ $\forall x \in \mathbb{R}$. Then also $cd \neq 0$ (because $cd = 0 \Rightarrow c^{-1}(cd) = c^{-1} \cdot 0 \Rightarrow d = 0$) and $f(x) = cg(x) = cdh(x)$ $\forall x \in \mathbb{R}$. Hence $f \sim h$.

Thus $\sim$ is an equivalence relation.

   (b) What is the equivalence class of the function $h: \mathbb{R} \to \mathbb{R}$ where $h(x) = 1$ for all $x \in \mathbb{R}$?

We compute directly from the definition:

$$[h] = \{f \in U \mid f \sim h\}$$
$$= \{f \in U \mid \exists c \neq 0 \text{ such that } f(x) = ch(x) \ \forall x \in \mathbb{R}\}$$
$$= \{f \in U \mid \exists c \neq 0 \text{ such that } f(x) = c \ \forall x \in \mathbb{R}\}$$
$$= \{f \in U \mid f \text{ is constant and nonzero}\}.$$

   (c) What is the equivalence class of the function $z: \mathbb{R} \to \mathbb{R}$ where $z(x) = 0$ for all $x \in \mathbb{R}$?

Again, we compute directly

$$[z] = \{f \in U \mid f \sim z\}$$
$$= \{f \in U \mid \exists c \neq 0 \text{ such that } f(x) = cz(x) \ \forall x \in \mathbb{R}\}$$
$$= \{f \in U \mid \exists c \neq 0 \text{ such that } f(x) = 0 \ \forall x \in \mathbb{R}\}$$
$$= \{f \in U \mid f(x) = 0 \ \forall x \in \mathbb{R}\}.$$
$$= \{z\}.$$

□

6. (10 points) Give a bijection $\mathbb{N} \to \mathbb{Z}$.

   Hint: Map even naturals to positive integers and odd naturals to non-positive integers.

We define $f: \mathbb{N} \to \mathbb{Z}$ by $f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd} \end{cases}$

First, we need to show that the codomain is actually $\mathbb{Z}$.

If $n \in \mathbb{N}$ is even, then $n = 2k$ for some $k \in \mathbb{Z}$, so

$f(n) = \frac{n}{2} = \frac{2k}{2} = k \in \mathbb{Z}$

If $n \in \mathbb{N}$ is odd. Then $n = 2k+1$ for some $k \in \mathbb{Z}$. So

$f(n) = -\frac{n-1}{2} = -\frac{2k+1-1}{2} = -k \in \mathbb{Z}$.

In any case, we see that $f(n) \in \mathbb{Z} \quad \forall n \in \mathbb{N}$

Now we show that f is injective and surjective.

Injectivity: Let $n, m \in \mathbb{N}$ with $f(n) = f(m)$. We show that $n = m$

We consider all possible cases,

Case 1: n even, m odd. In this case $f(n) = \frac{n}{2} > 0$ by positive closure

and $f(m) = -\frac{m-1}{2} \leq 0$, so we can't have $f(n) = f(m)$ in this case.

Case 2: n odd, m even. Works the same way as previous case.

Case 3: n even, m even. In this case $f(n) = f(m)$ says that $\frac{n}{2} = \frac{m}{2}$,

and multiplying by 2 we get $n = m$ as desired.

Case 4: n odd, m odd. Now the assumption $f(n) = f(m)$ gives

$f(n) = f(m) \Rightarrow -\frac{n-1}{2} = -\frac{m-1}{2} \Rightarrow \frac{n-1}{2} = \frac{m-1}{2} \Rightarrow n-1 = m-1 \Rightarrow n = m$

In any case, we see that $f(n) = f(m)$ is only possible for $n = m$, so f injective.

Surjectivity: Take an arbitrary $k \in \mathbb{Z}$. We show that there exists some $n \in \mathbb{N}$ such that $f(n) = k$.

Case 1: $k = 0$. In this case $f(1) = -\frac{1-1}{2} = 0$ ($n=1$ works)

Case 2: $k > 0$. Here we see that $n = 2k \in \mathbb{N}$ satisfies

$$f(n) = f(2k) \underset{\downarrow 2k \text{ even}}{=} \frac{2k}{2} = k \text{ as desired}$$

Case 3: $k < 0$. Now we have $-k > 0$ so $n = -2k+1 \in \mathbb{N}$ and

$$f(n) = f(-2k+1) \underset{\downarrow -2k+1 \text{ odd}}{=} -\frac{-2k+1-1}{2} = -\frac{-2k}{2} = k \quad \text{as desired.}$$

Thus f is surjective

It follows that f is bijective $\square$

Final Exam Notesheet

Divisibility. For $m, n \in \mathbb{Z}$ with $m \neq 0$, m *divides* n if $n = mq$ for some $q \in \mathbb{Z}$. Denoted $m \mid n$.

(D1) Divisibility is reflexive and transitive.

(D2) If $a \mid b$ then $a \mid bc$.

(D3) The Division Theorem. For $a, b \in \mathbb{Z}$ with $b > 0$, there exists unique $q, r \in \mathbb{Z}$ with $a = bq + r$ and $0 \leqslant r < b$.

Congruence. For $n, a, b \in \mathbb{Z}$ with $n > 1$, we say $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

(C1) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $a + c \equiv b + d \pmod{n}$.

(C2) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $ac \equiv bd \pmod{n}$.

(C3) If $a \equiv b \pmod{n}$ then $a^m \equiv b^m \pmod{n}$ for any $m \in \mathbb{N}$.

(C4) For any $a \in \mathbb{Z}$ there is a unique $r \in \mathbb{Z}$ with $0 \leqslant r < n$ such that $a \equiv r \pmod{n}$.

Sets. Let A, B and C be subsets of some universal set U.

(S1) $A \subseteq B$ when $x \in A \Rightarrow x \in B$.

(S2) $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.

(S3) $\mathcal{P}(A) = \{D \subseteq U \mid D \subseteq A\}$.

(S4) If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$ for any $n \geqslant 0$.

(S5) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(S6) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

(S7) $(A^c)^c = A$ and $A \backslash B = A \cap B^c$.

(S8) $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$.

Indexed Families of Sets. Let Λ be a not empty (index) set. Suppose that we have a set A_α for each $\alpha \in \Lambda$.

(IFS1) Union: $\bigcup_{\alpha \in \Lambda} A_\alpha = \{x \mid x \in A_\alpha \text{ for some } \alpha \in \Lambda\}$. Intersection: $\bigcap_{\alpha \in \Lambda} A_\alpha = \{x \mid x \in A_\alpha \text{ for every } \alpha \in \Lambda\}$.

(IFS3) For any $\beta \in \Lambda$, we have $A_\beta \subseteq \bigcup_{\alpha \in \Lambda} A_\alpha$ and $\bigcap_{\alpha \in \Lambda} A_\alpha \subseteq A_\beta$.

(IFS4) De Morgan's Laws: $\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$ and $\left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$.

Functions. Let A and B be non-empty sets.

(F1) A function $f\colon A \to B$ is injective if, for any $x, y \in A$, $f(x) = f(y) \Rightarrow x = y$.

(F2) A function $f\colon A \to B$ is surjective if for every b in B there exists some $a \in A$ such that $f(a) = b$.

(F3) A function is bijective if it is injective and surjective.

(F4) If $V \subseteq A$ we denote $f(V) = \{f(x) \mid x \in V\} \subseteq B$. If $Y \subseteq B$, we denote $f^{-1}(Y) = \{x \in A \mid f(x) \in Y\} \subseteq A$.

(F5) If $f\colon A \to B$ is a function we denote $f^{-1} = \{(f(a), a) \in B \times A \mid a \in A\}$.

(F6) If $f\colon A \to B$ is bijective then f^{-1} defines a function $f^{-1}\colon B \to A$ defied by $f^{-1}(b) = a \Leftrightarrow f(a) = b$.

(F7) If $f\colon A \to B$ and $g\colon B \to C$, the composition $g \circ f\colon A \to C$ is defined by $(g \circ f)(x) = g(f(x))$.

Relations. Let A and B be sets. A relation R from A to B is a subset $R \subset A \times B$. We denote $a \sim b$ to denote $(a, b) \in R$. A relation on a set A is a relation from A to A.

(R1) A relation $\sim$ on A is an equivalence relation if it is:

- Reflexive: $x \sim x$ for all $x \in A$.
- Symmetric $x \sim y \Rightarrow y \sim x$.
- Transi: ($x \sim y$ and $y \sim z$) $\Rightarrow x \sim z$.

(R2) If $\sim$ is an equivalence relation, the equivalence class of $a \in A$ is $[a] = \{x \in A\colon x \sim a\}$.

(R3) For $n \in \mathbb{N}$, we denote by $\mathbb{Z}_n$ the set of equivalence classes for the relation of congruence modulo n.

Equinumerousity. Two sets A and B are equinumerous if there is a bijection $A \to B$. Denoted $A \approx B$.

(E1) Equinumerousity is an equivalence relation.

(E2) A is finite if either $A = \varnothing$ or $A \approx \mathbb{N}_k = \{1, 2, \ldots, k\}$ for some $k \in \mathbb{N}$.

(E3) If B is finite and $A \subseteq B$ then A is finite and $\text{card}(A) \leqslant \text{card}(B)$.

(E4) If A is equinumerous to one of its proper subsets, then A is infinite.

(E5) Pigeonhole: If A and B are finite and $\text{card}(A) > \text{card}(B)$, there is no injection $A \to B$.

(E6) Surjective Pigeonhole: If A and B are finite and $\text{card}(A) < \text{card}(B)$, there is no surjection $A \to B$.

(E6) A set A is countably infinite if $A \approx \mathbb{N}$. The sets $\mathbb{Z}$, $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are countably infinite.

(E7) A set is countable if it is either finite or countably infinite.

(E8) A set is uncountable if it is not countable. The open interval $(0, 1)$ is uncountable.

(E9) Any $a \in (0, 1)$ admits a unique decimal expression $a = 0.a_1 a_2 a_3 \ldots$ not ending in an infinite string of 9's.